

TE Curve Verification Pipeline Familywise ONNX Report - lstm_sequence

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/lstm_sequence/2026-07-08-23-29-14__track2_lstm_sequence_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/lstm_sequence/2026-07-08-23-29-14__track2_lstm_sequence_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV: `output/validation_checks/track2_familywise_onnx_report/lstm_sequence/2026-07-08-23-29-14__track2_lstm_sequence_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV: `output/validation_checks/track2_familywise_onnx_report/lstm_sequence/2026-07-08-23-29-14__track2_lstm_sequence_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `lstm_sequence` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_lstm_sequence_fw_simplified_setpoints</code>	<code>2026-07-08-15-14-13_te_lstm_sequence_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_lstm_sequence_bw_simplified_setpoints</code>	<code>2026-07-08-15-24-03_te_lstm_sequence_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_lstm_sequence_global_simplified_setpoints</code>	<code>2026-07-08-14-59-31_te_lstm_sequence_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/forward/2026-07-08-15-14-13_te_lstm_sequence_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/forward/2026-07-08-15-14-13_te_lstm_sequence_fw_simplified_setpoints/python/lstm_sequence-epoch=092-val_mae=0.00370236.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/backward/2026-07-08-15-24-03_te_lstm_sequence_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/backward/2026-07-08-15-24-03_te_lstm_sequence_bw_simplified_setpoints/python/lstm_sequence-epoch=091-val_mae=0.00367749.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/global/2026-07-08-14-59-31_te_lstm_sequence_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/lstm_sequence/global/2026-07-08-14-59-31_te_lstm_sequence_global_simplified_setpoints/python/lstm_sequence-epoch=115-val_mae=0.00369210.ckpt</code>

Aggregate Metrics

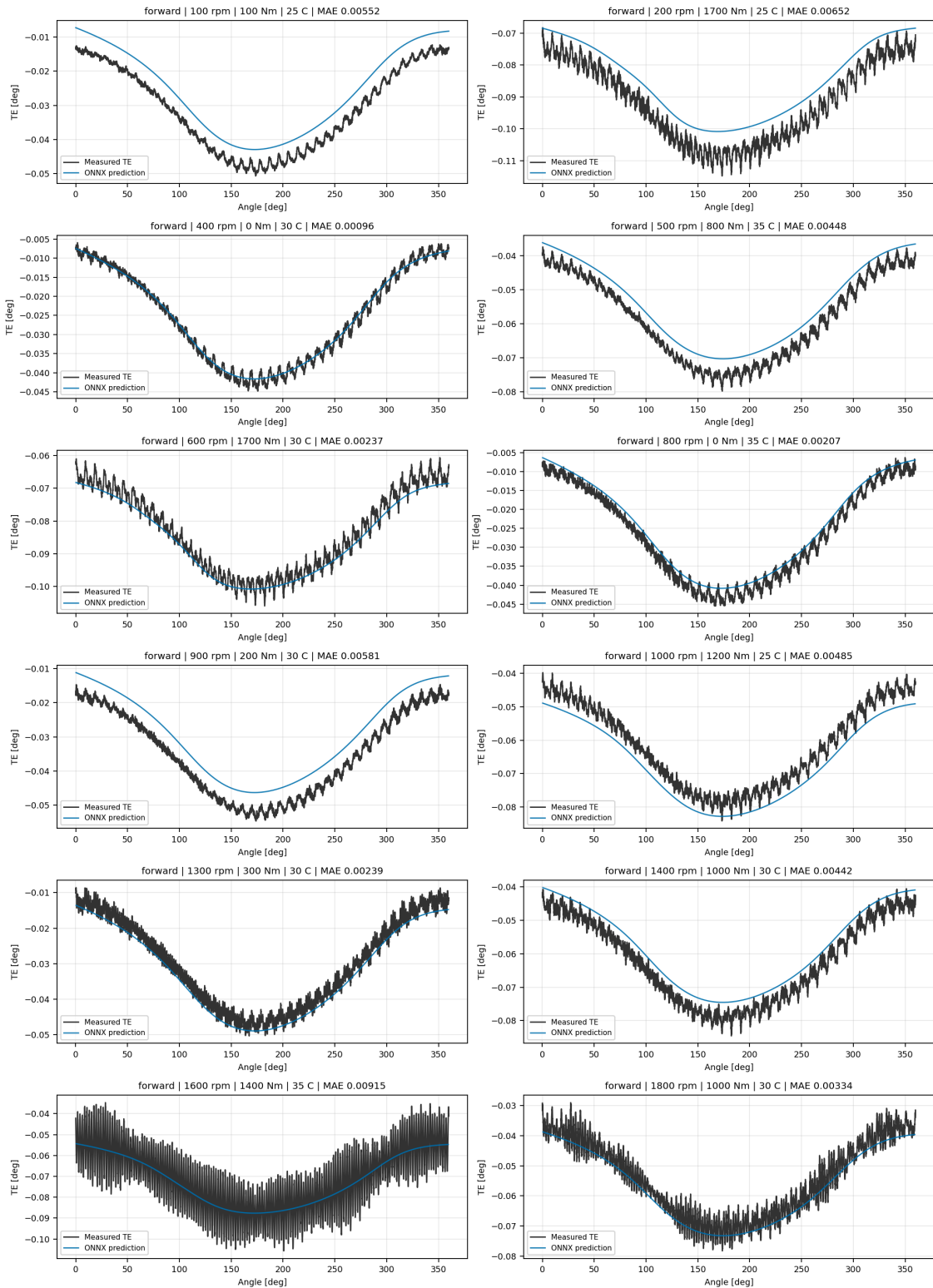
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003491	0.003916	8.190	15.462
backward	97	0.003584	0.004038	8.305	15.374
global	194	0.003446	0.003867	8.014	13.715

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000662	0.002947	0.001703	0.008429
backward	0.000626	0.002837	0.001903	0.007710
global	0.000386	0.002792	0.001776	0.007126

Forward 12-Curve Page

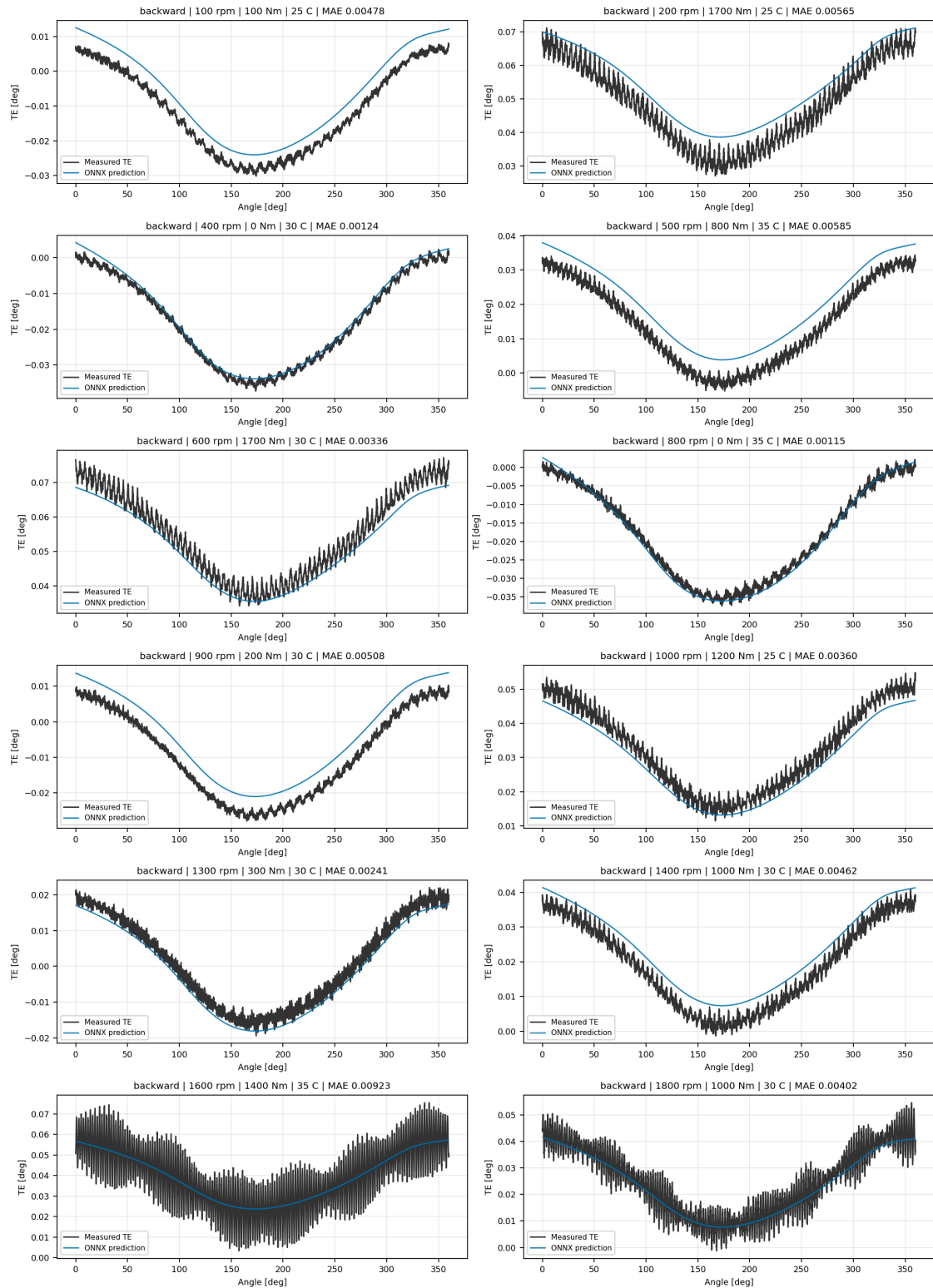
Simplified Dataset + Setpoints | lstm_sequence | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

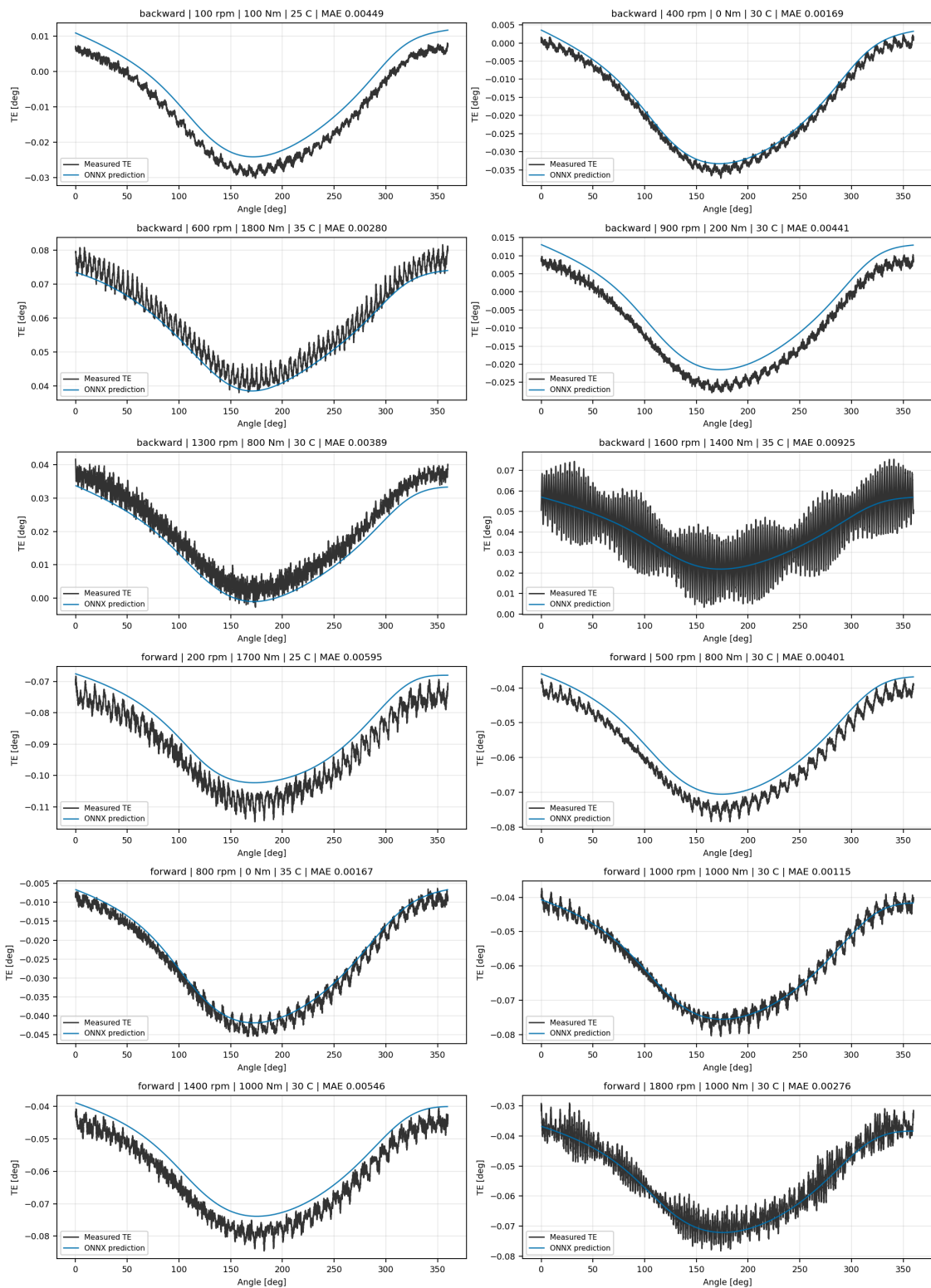
Simplified Dataset + Setpoints | lstm_sequence | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | lstm_sequence | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: lstm_sequence ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_lstm_sequence_fw_polished_setpoints	2026-07-08-16-05-59_te_lstm_sequence_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_lstm_sequence_bw_polished_setpoints	2026-07-08-16-21-04_te_lstm_sequence_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_lstm_sequence_global_polished_setpoints	2026-07-08-15-45-54_te_lstm_sequence_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/lstm_sequence/forward/2026-07-08-16-05-59_te_lstm_sequence_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/lstm_sequence/forward/2026-07-08-16-05-59_te_lstm_sequence_fw_polished_setpoints/python/lstm_sequence-epoch=078-val_mae=0.00219094.ckpt
backward	models/polished_dataset/setpoints/exported/lstm_sequence/backward/2026-07-08-16-21-04_te_lstm_sequence_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/lstm_sequence/backward/2026-07-08-16-21-04_te_lstm_sequence_bw_polished_setpoints/python/lstm_sequence-epoch=070-val_mae=0.00219971.ckpt
global	models/polished_dataset/setpoints/exported/lstm_sequence/global/2026-07-08-15-45-54_te_lstm_sequence_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/lstm_sequence/global/2026-07-08-15-45-54_te_lstm_sequence_global_polished_setpoints/python/lstm_sequence-epoch=118-val_mae=0.00218648.ckpt

Aggregate Metrics

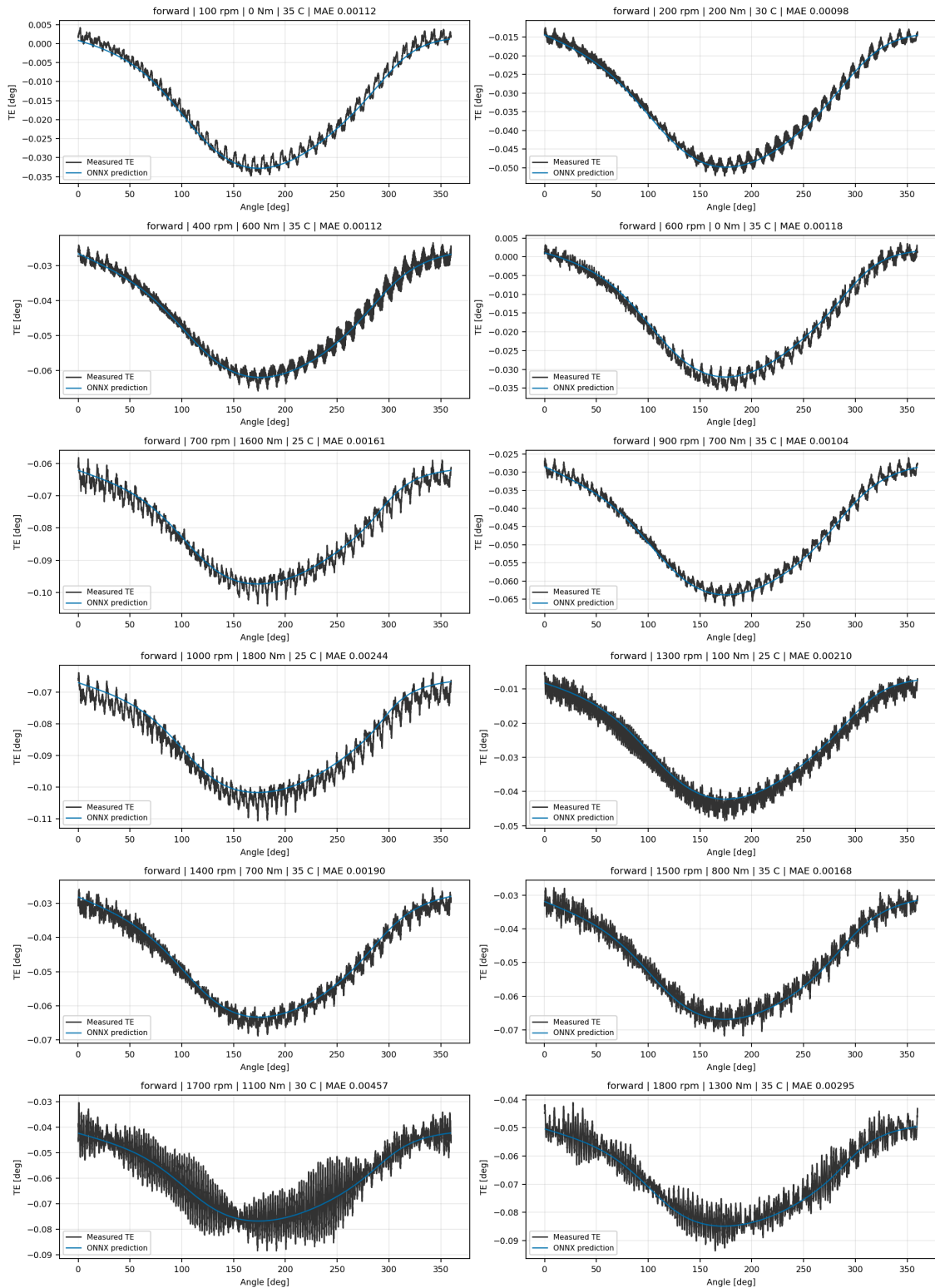
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002222	0.002682	4.922	9.879
backward	94	0.002739	0.003246	5.110	11.311
global	194	0.002470	0.002950	4.999	11.303

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000290	0.000991	0.001905	0.007869
backward	0.000011	0.000958	0.002377	0.008768
global	0.000065	0.001013	0.002136	0.008355

Forward 12-Curve Page

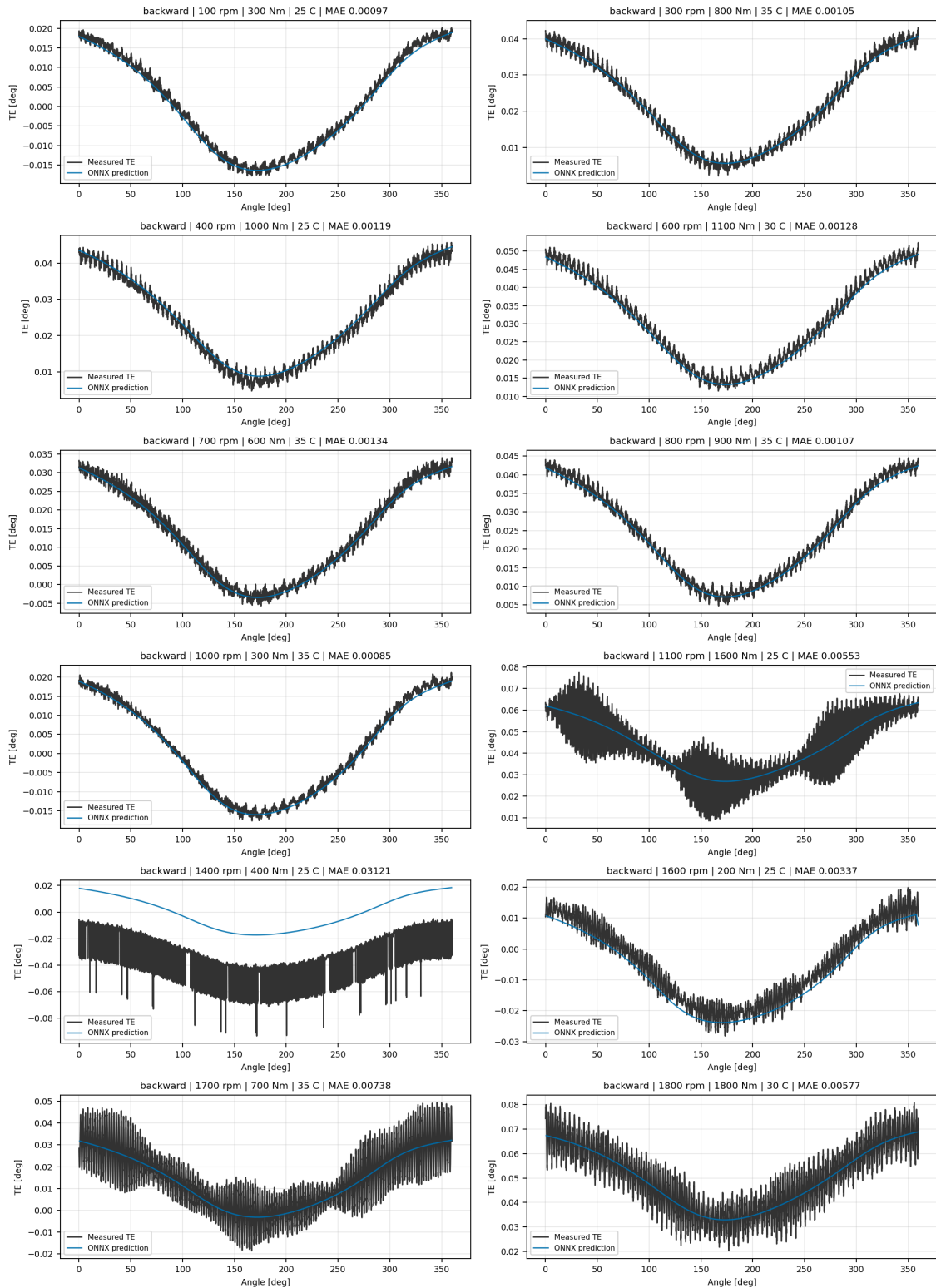
Polished Dataset + Setpoints | lstm_sequence | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

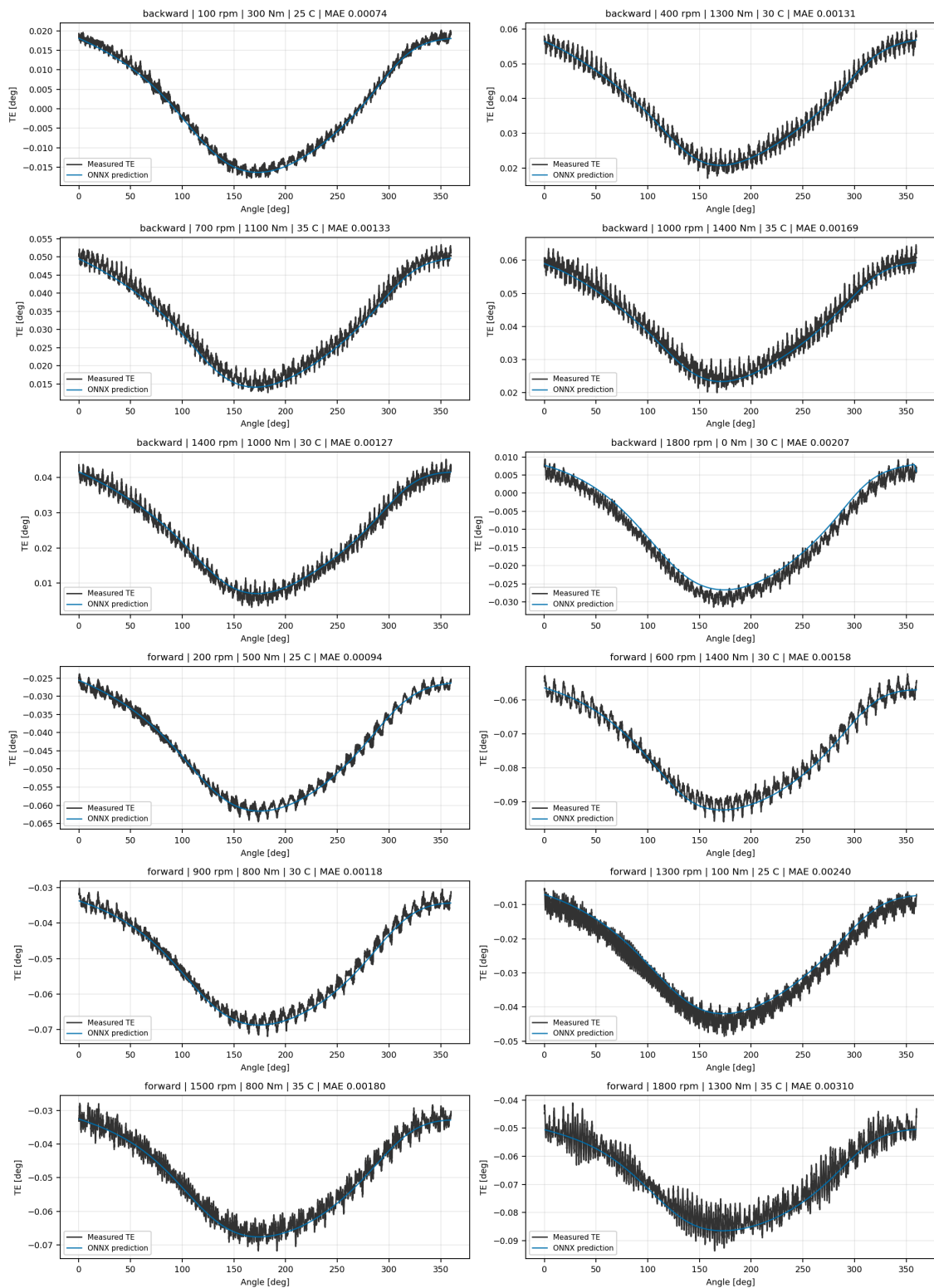
Polished Dataset + Setpoints | lstm_sequence | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | lstm_sequence | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: lstm_sequence ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_lstm_sequence_fw_polished_actua l_values	2026-07-08-17-20- 19_te_lstm_sequence_fw_polished_ac tual_values	polished_point_v1
backward	te_lstm_sequence_bw_polished_actua l_values	2026-07-08-18-02- 26_te_lstm_sequence_bw_polished_ac tual_values	polished_point_v1
global	te_lstm_sequence_global_polished_a ctual_values	2026-07-08-16-51- 13_te_lstm_sequence_global_polishe d_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/l stm_sequence/forward/2026-07-08-17-20- 19_te_lstm_sequence_fw_polished_actual_values/o nnx/model.onnx	models/polished_dataset/actual_values/exported/l stm_sequence/forward/2026-07-08-17-20- 19_te_lstm_sequence_fw_polished_actual_values/p ython/lstm_sequence-epoch=210- val_mae=0.00215124.ckpt
backward	models/polished_dataset/actual_values/exported/l stm_sequence/backward/2026-07-08-18-02- 26_te_lstm_sequence_bw_polished_actual_values/o nnx/model.onnx	models/polished_dataset/actual_values/exported/l stm_sequence/backward/2026-07-08-18-02- 26_te_lstm_sequence_bw_polished_actual_values/p ython/lstm_sequence-epoch=240- val_mae=0.00214547.ckpt
global	models/polished_dataset/actual_values/exported/l stm_sequence/global/2026-07-08-16-51- 13_te_lstm_sequence_global_polished_actual_valu es/onnx/model.onnx	models/polished_dataset/actual_values/exported/l stm_sequence/global/2026-07-08-16-51- 13_te_lstm_sequence_global_polished_actual_valu es/python/lstm_sequence-epoch=160- val_mae=0.00218895.ckpt

Aggregate Metrics

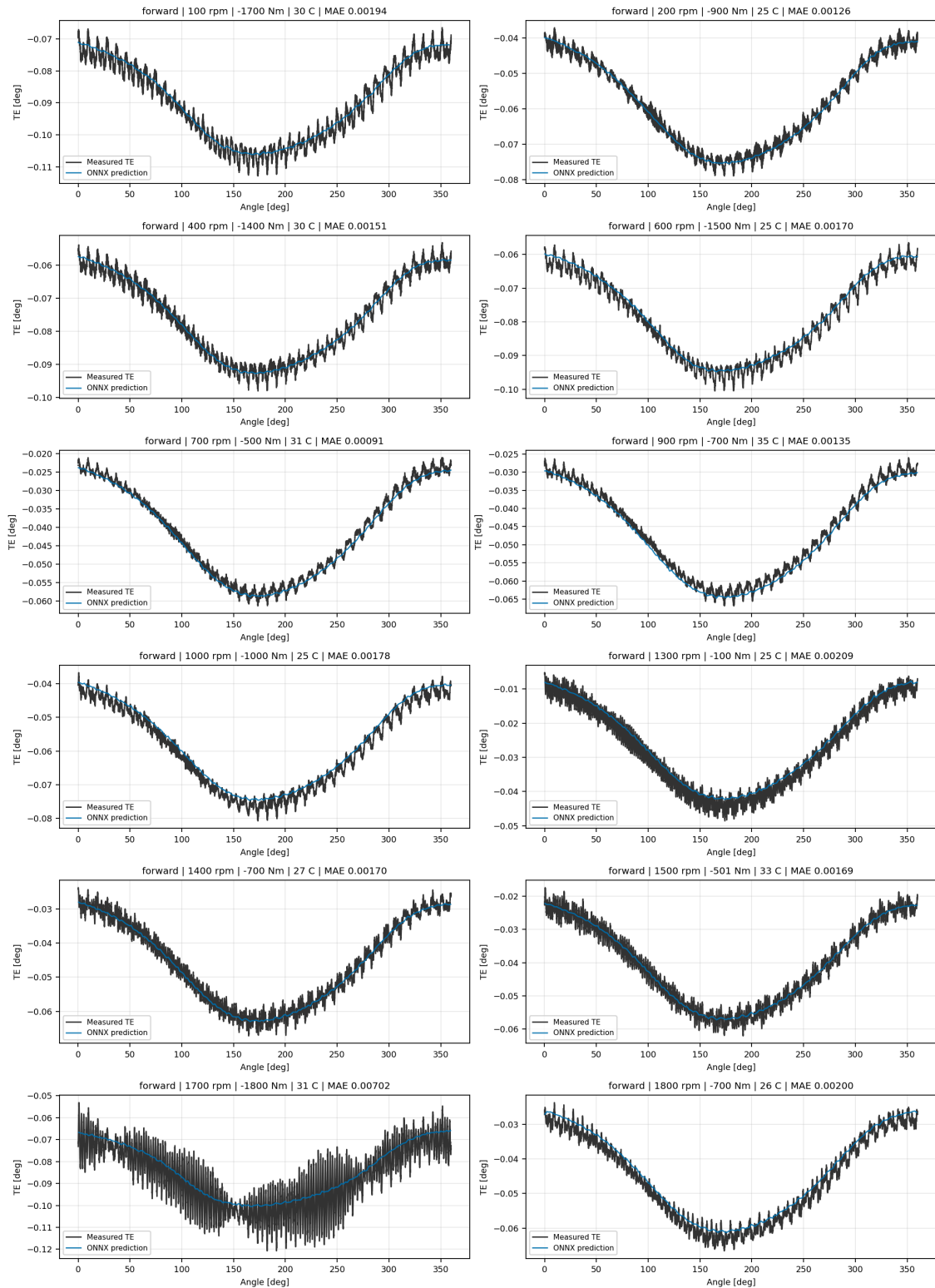
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002071	0.002521	4.543	9.394
backward	94	0.002424	0.002946	4.688	11.020
global	194	0.002281	0.002770	4.710	10.519

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000058	0.000707	0.001914	0.007964
backward	-0.000063	0.000527	0.002352	0.008308
global	0.000144	0.000685	0.002126	0.007977

Forward 12-Curve Page

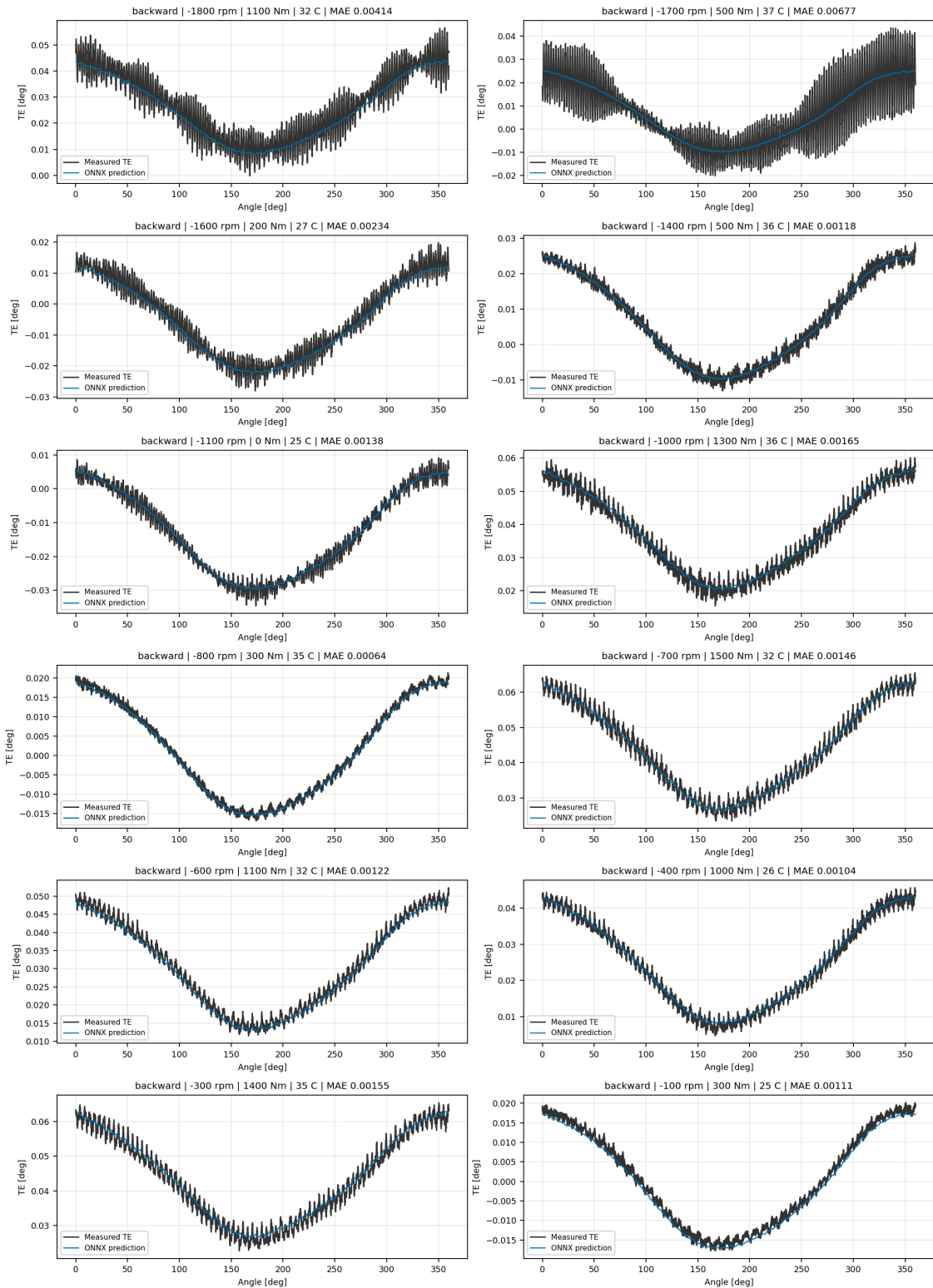
Polished Dataset + Actual Values | lstm_sequence | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

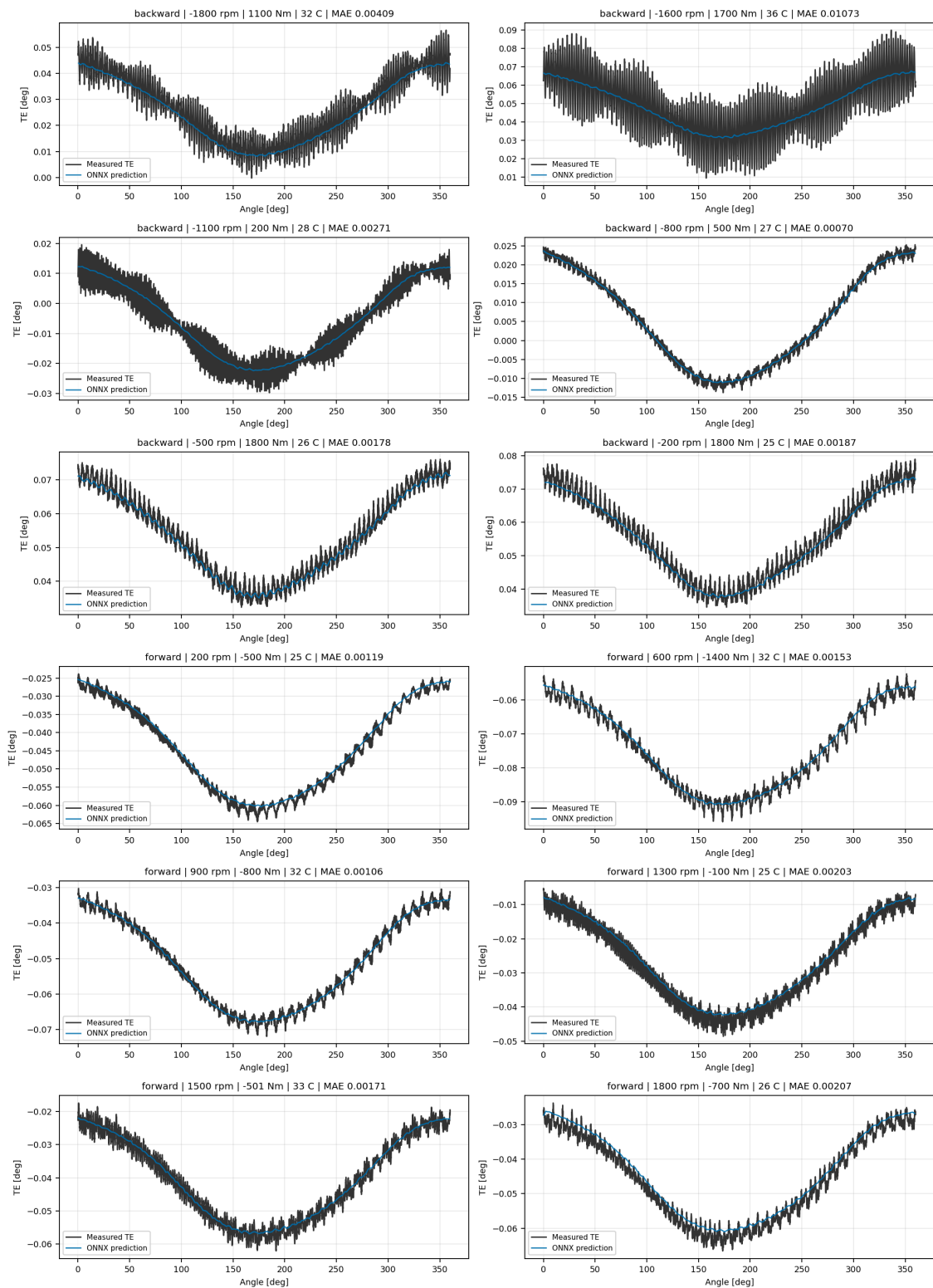
Polished Dataset + Actual Values | lstm_sequence | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | lstm_sequence | global



polished_dataset_actual_values global 12-curve collage